Supplementary Information

CrossPiezo: Limited Elite-Set Reproducibility in Cross-Database Piezoelectric Screening

Anonymous Author(s)

Contents

1 Supplementary notes

1.1 Portfolio estimand definition (Note 1)

The portfolio benchmark reports two related quantities. *Worst-source recall* is a material-level (micro) metric: it counts how many of the top-q^*n materials selected by a strategy are also present in the JARVIS and MP elite sets, divided by q^*n, with no group weighting. *Paired difference in recall* is the difference between the material-level worst-source recall of the strategy and that of the better single-source baseline, computed on the full observed panel.

The confidence interval reported in Table S3 uses a *full-procedure grouped bootstrap*. Each replicate: (i) resamples reduced-formula groups with replacement and assigns distinct bootstrap identities to duplicated occurrences; (ii) recomputes source-specific rankings and top-q^* elite sets on the bootstrap sample; (iii) re-runs the portfolio strategy and both single-source baselines; (iv) computes the improvement over the better single source within that replicate. The reported 95% interval is the percentile bootstrap interval of these replicate differences. The point estimate is fixed on the full-panel selections and is exactly the arithmetic difference of the Recall column; the bootstrap captures uncertainty from group resampling and re-selection.

The "better single source" is determined per replicate for the interval; the main-text point estimate uses the full-panel better source (here JARVIS-only and MP-only are tied on P0 F1 at $q^* = 10\%$, $b = 1.0$). The portfolio analysis is presented as a risk-management illustration, not as a validated method comparison or as evidence of physical truth.

1.2 Structure matching and manual audit (Note 2)

The CrossPiezo-Invariant-v1 panel was constructed from relaxed CIFs shared by the Materials Project (MP) and JARVIS. The matching pipeline is: (i) load all processed MP and JARVIS piezoelectric records; (ii) compute structure-level overlaps by compositional reduced formula and relaxed-lattice similarity; (iii) apply a frozen similarity threshold to obtain candidate pairs; (iv) perform a response-blind manual audit of the top-matched candidates using a frozen rule set. The audit was response-blind: the auditor did not see piezoelectric response values when deciding whether a candidate pair represented the same or a distinct structure. All audit decisions, exclusion reasons and ambiguous cases are recorded in the frozen panel-membership file.

The frozen matcher tolerances are: fractional length tolerance $\ell_{\mathrm{tol}} = 0.2$, site tolerance $s_{\mathrm{tol}} = 0.3\,\text{Å}$, angle tolerance $5°$, primitive-cell reduction enabled, and lattice scaling enabled (see "configs/matching.yaml"). P0 ($n = 573$) is the full set of accepted structure matches; P2 ($n = 207$) is the subset passing tighter RMS-distance and lattice-distance thresholds ($\mathrm{RMS_{thr}} \approx 0.0078\,\text{Å}$, lattice-distance threshold ≈ 0.0036). Table S6 gives the resulting distance distributions. Figure S5 shows an association that is statistically detectable at this panel size but negligible in magnitude (Spearman $\rho = -0.09$, $p = 0.0312$); this is a sensitivity diagnostic rather than an exclusion of threshold or matching-error effects.

1.3 Tensor invariants and conventions (Note 3)

All piezoelectric response values are piezoelectric stress tensors $\mathbf{e}$ in $\mathrm{C/m^2}$ with full Cartesian $3 \times 3 \times 3$ components. Using the orthonormal Mandel representation

$$M = \begin{bmatrix} e_{1xx} & e_{1yy} & e_{1zz} & \sqrt{2}e_{1yz} & \sqrt{2}e_{1xz} & \sqrt{2}e_{1xy} \\ e_{2xx} & e_{2yy} & e_{2zz} & \sqrt{2}e_{2yz} & \sqrt{2}e_{2xz} & \sqrt{2}e_{2xy} \\ e_{3xx} & e_{3yy} & e_{3zz} & \sqrt{2}e_{3yz} & \sqrt{2}e_{3xz} & \sqrt{2}e_{3xy} \end{bmatrix},$$

and $m(\mathbf{n}) = (n_x^2, n_y^2, n_z^2, \sqrt{2}n_y n_z, \sqrt{2}n_x n_z, \sqrt{2}n_x n_y)^\mathsf{T}$ for $\|\mathbf{n}\|_2 = 1$, we define

$$F_1 = \|M\|_F, \qquad F_4 = \|M\|_2 = \sigma_{\max}(M), \qquad F_3 = \max_{\|\mathbf{n}\|_2=1} |\mathbf{n}^\mathsf{T} M m(\mathbf{n})|.$$

The internal check $0 \le F_3 \le F_4 \le F_1$ follows from these definitions. The deterministic F3 implementation uses a 10,000-point Fibonacci sphere, 20 projected-gradient starts, at most 100 iterations per start with improvement tolerance 10^{-9}, and a bounded Nelder–Mead spherical-coordinate polish with 400 iterations and objective tolerances $10^{-9}/10^{-12}$; it is cross-checked against a dense brute-force oracle in the convention audit. Voigt-to-Cartesian conversion follows the standard convention $11 \to xx$, $22 \to yy$, $33 \to zz$, $23 \to yz$, $13 \to xz$, $12 \to xy$ with factor-of-two handling for shear components as appropriate for the stress tensor. No sign conventions, unit conversions or coordinate rotations were applied silently; all transformations are recorded in the repository transformation history.

1.4 Bootstrap, quantile and tie handling (Note 4)

For equations below, q is written as a fraction; the reported grid is $q = 1\%, 2\%, \ldots, 50\%$. For a panel of size N, let $k(q) = \max\{1, \lfloor qN \rfloor\}$ and let $T_A(q)$ and $T_B(q)$ be the two top-$k(q)$ sets. With $X_q = |T_A(q) \cap T_B(q)|$, $J_q = X_q/(2k(q) - X_q)$. Under independent random rankings, $X_q \sim \text{Hypergeom}(N, k, k)$ and

$$E_0[J_q] = \sum_{x=\max(0,2k-N)}^{k} \frac{x}{2k - x} \frac{\binom{k}{x}\binom{N-k}{k-x}}{\binom{N}{k}},$$

not J_q evaluated at $E[X_q]$. The chance-adjusted overlap is $\widetilde{J}_q = (J_q - E_0[J_q])/(1 - E_0[J_q])$; it can be negative, so nAUCC is not restricted to $[0, 1]$. For the primary cluster-bootstrap analysis, curve and onset inference use one reduced-formula cluster bootstrap with a studentized sup-norm critical value; scalar nAUCC and partial nAUCC intervals are percentile intervals from the same replicates. Each replicate resamples reduced-formula groups with replacement, recomputes the bootstrap universe size and exact hypergeometric null, and then recomputes the complete screening curve.

For property-control comparisons and the portfolio benchmark, bootstrap resampling is also by reduced-formula group to respect the dependence among entries sharing a composition. In the portfolio full-procedure bootstrap, duplicated groups create distinct bootstrap identities so that re-selection and elite-set sizes are computed on the expanded bootstrap sample. The random seed, number of replicates, studentization details, tie handling and k rounding are recorded in the configuration and result files. Ties in source rankings are broken by stable sorting; duplicate reduced formulae are retained in the ranking but noted in the panel metadata. We distinguish two onset concepts. The persistent above-chance onset is $q_0^{\text{persist}} = \min\{q_j : \text{LCB}(q_{j+r}) > 0,\ r = 0, \ldots, L-1\}$, whereas the prespecified persistent practical-onset threshold is $q_{0.05}^{\text{persist}} = \min\{q_j : \text{LCB}(q_{j+r}) > 0.05,\ r = 0, \ldots, L-1\}$. We use $L = 5$ and report $q_{0.05}^{\text{persist}}$ as the primary practical threshold; 0.05 is a prespecified practical threshold, not the unique criterion for statistical distinguishability from chance. The legacy row-bootstrap curve is retained as a sensitivity analysis. Tables S7a and S7b report the primary cluster-bootstrap results; the tie-aware cutoff audit additionally reports exact overlap bounds when a score tie straddles k, together with absolute and relative score gaps. No such ambiguity occurs at the P0 F1 cutoffs $q = 1\%, 5\%, 10\%$.

1.5 Source workflow and tensor conventions (Note 5)

Both JARVIS and MP report the piezoelectric stress tensor $\mathbf{e}$ in C/m^2 using the same Voigt order (xx, yy, zz, yz, xz, xy) and the same engineering-shear factor-of-two convention. JARVIS values are expressed in the source-structure Cartesian frame; MP values are expressed in the MP IEEE-oriented structure frame. Before any response comparison, MP structures are aligned to JARVIS structures (or vice versa) with a frozen "pymatgen" "StructureMatcher" configuration and the recovered rotation is applied to the MP Cartesian tensor. The three reported scalars (F1, F3, F4) are coordinate-frame invariants, so they are unaffected by the source-specific structure-frame choice.

Several calculation settings are not documented in the release metadata used by CrossPiezo: exchange–correlation functional, pseudopotential family, DFPT vs finite-difference ionic response, plane-wave cutoff, k-point density, relaxation thresholds, primitive vs conventional defaults, and sign-convention history. These items are therefore reported as unknown rather than assumed identical (see `docs/mp_vs_jarvis_conventions.md`). The repository convention audit (`scripts/run_convention_audit.py`) confirms that F1, F3 and F4 are rotation invariant, that Voigt/Cartesian conversion and engineering-shear handling are internally consistent, and that the F3 optimiser converges to a dense-grid brute-force oracle. This audit rules out low-level convention errors in the CrossPiezo pipeline; it does not prove that JARVIS and MP use identical calculation settings.

1.6 Screening resolution vs conventional ranking diagnostics (Note 6)

Screening resolution is reported with the chance-adjusted Jaccard curve $\widetilde{J}_q$ and its area (nAUCC). These metrics focus on the elite tail, where cross-database disagreement is largest. For comparison, Table S5 and Figures S3–S4 report conventional diagnostics: plain Jaccard J_q, chance-adjusted Jaccard $\widetilde{J}_q$, precision@k (equal here to recall@k and the overlap coefficient), Rank-Biased Overlap (RBO) with $p = 0.95$, top-weighted Kendall τ on the union of the two top-k sets, and top-weighted Spearman ρ on the same union.

These diagnostics illustrate that global rank association and elite-set agreement answer different questions. The synthetic example in Figure S3 constructs two length-100 rankings with global Kendall $\tau \approx 0.98$ but disjoint top-5% sets; global metrics are near-perfect while top-5% precision and Jaccard are zero. On P0 F1, global Kendall $\tau = 0.245$ and Spearman $\rho = 0.350$ are weak-to-moderate rather than high; elite-tail agreement is nevertheless near chance for small q (see Table S5).

2 Supplementary figures

3 Supplementary tables

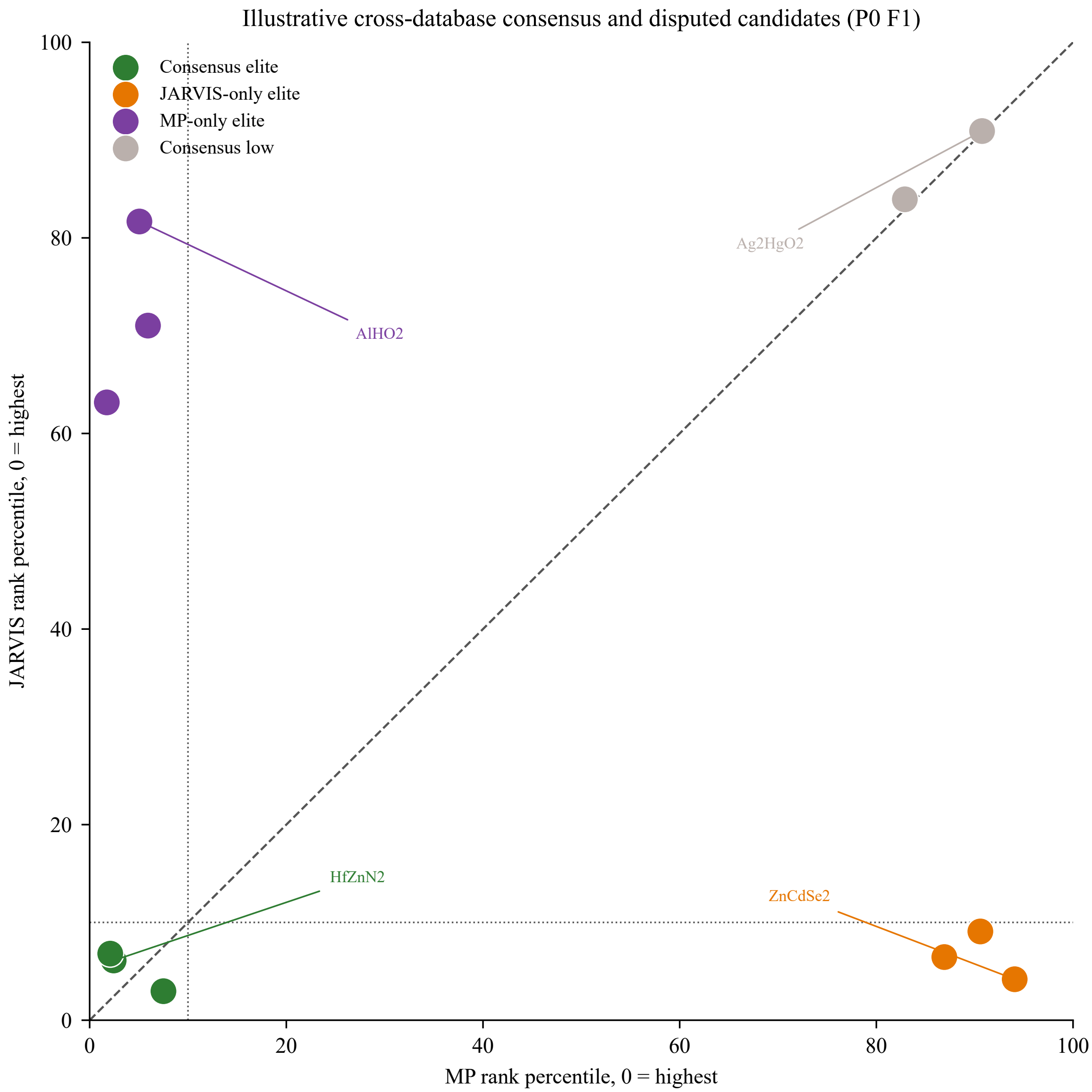

Figure S1: Illustrative cross-database consensus and disputed candidates on P0 F1. Points show JARVIS versus MP rank percentiles (0 = highest) for consensus elite (green), JARVIS-only elite (orange), MP-only elite (purple) and consensus low (grey) materials. The dashed diagonal marks equal ranking; dotted lines mark the 10% elite threshold. These examples are illustrative and are not adjudicated as physically correct or incorrect.

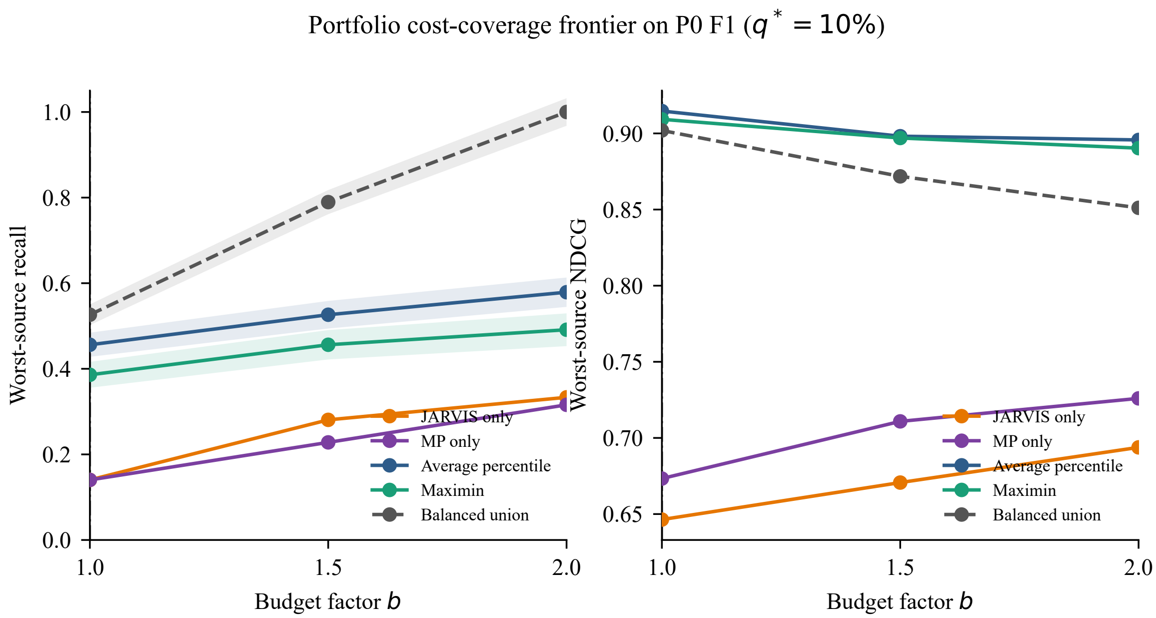

Figure S2: Portfolio cost–coverage frontier on P0 F1 ($q^* = 10\%$). Worst-source recall (left) and NDCG (right) are plotted against budget factor b for the best single-source baselines and three source-robust aggregation strategies. The vertical dashed line marks the equal-budget analysis at $b = 1.0$. Aggregation changes the coverage–quality trade-off but does not validate physical truth. See Note 1 for the distinction between material-level recall and group-mean paired differences.

Table S1: Corrected high-response sensitivity for P0 F1 at $f = 0.10$ and $f = 0.25$.

Method	f	N	τ	CI	nAUCC	CI
Dual-high	10%	58	0.035	[-0.193, 0.239]	-0.046	[-0.118, 0.044]
JARVIS anchor	10%	57	0.209	[-0.062, 0.374]	0.053	[-0.061, 0.191]
MP anchor	10%	57	-0.053	[-0.235, 0.137]	-0.018	[-0.104, 0.090]
Pooled (exploratory)	10%	58	–	[–, –]	-0.155	[–, –]
Dual-high	25%	143	0.070	[-0.046, 0.181]	0.031	[-0.034, 0.102]
JARVIS anchor	25%	143	-0.019	[-0.141, 0.105]	-0.006	[-0.070, 0.070]
MP anchor	25%	143	0.039	[-0.083, 0.154]	0.007	[-0.052, 0.078]
Pooled (exploratory)	25%	144	–	[–, –]	-0.100	[–, –]

Table S2: Property-control comparison on P0. The high identical-value fraction flag marks a high fraction of identical cross-source values; for energy above hull this is largely associated with tied zero-hull entries. The flag does not establish copying or shared provenance. The $\Delta\tau$ and ΔnAUCC columns are full-panel point estimates.

Attribute	N	τ	95% CI	nAUCC	95% CI	$\Delta\tau$	ΔnAUCC	Identical-value flag
Volume	573	0.967	[0.963, 0.971]	0.918	[0.901, 0.936]	0.722	0.813	
Band gap	573	0.908	[0.890, 0.923]	0.897	[0.864, 0.908]	0.663	0.792	
Energy above hull	573	0.611	[0.547, 0.670]	0.533	[0.455, 0.592]	0.366	0.428	✓
Dielectric trace	573	0.482	[0.418, 0.547]	0.286	[0.228, 0.344]	0.236	0.181	

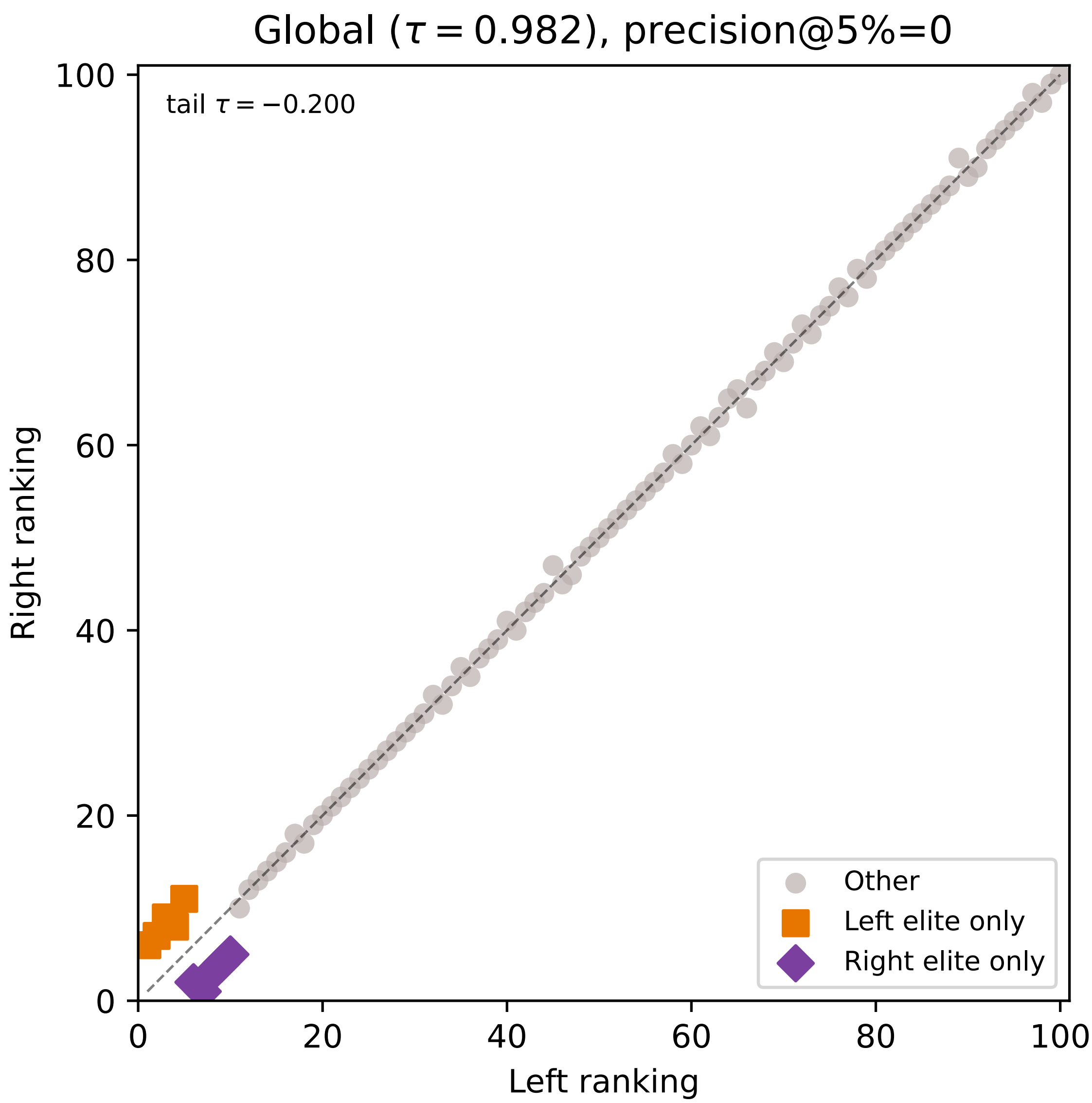

Figure S3: Synthetic motivating example for screening resolution. Two length-100 rankings are constructed with global Kendall $\tau \approx 0.98$ but disjoint top-5% sets, showing that high global rank correlation does not guarantee agreement on elite-screening decisions. The reported conventional metrics (global τ, Spearman ρ, Rank-Biased Overlap) are near-perfect while top-5% precision and Jaccard are zero. See Note 6 for the full metric definitions.

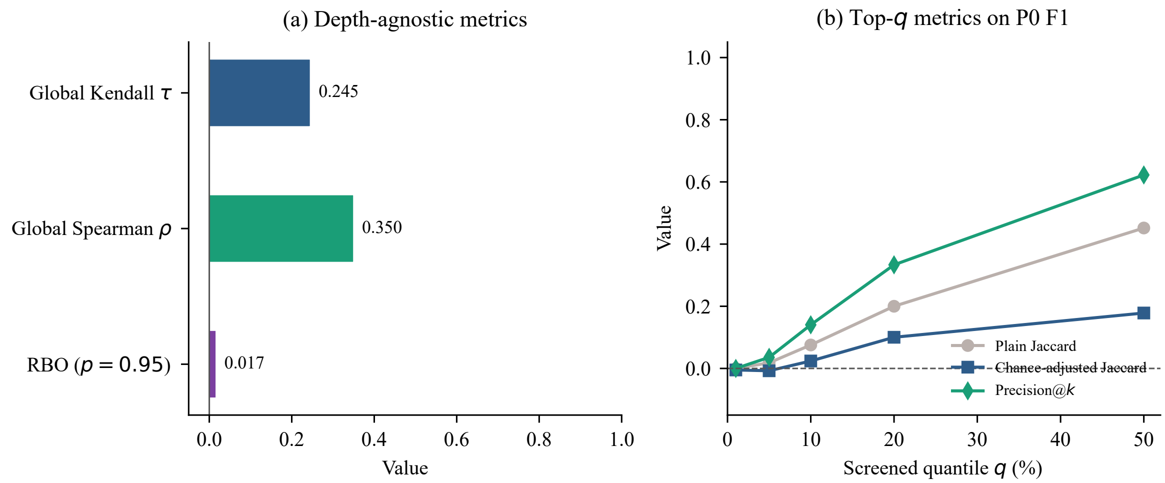

Figure S4: Comparison of conventional ranking diagnostics and screening-resolution on P0 F1. Left: depth-agnostic global Kendall τ, global Spearman ρ, and RBO. Right: set-based top-q diagnostics on P0 F1. Global rank association is modest, and elite-tail agreement is near-chance for small q; global association alone does not determine elite-set reproducibility. See Table S5 for the numerical values and Note 6 for definitions.

Table S3: Equal-budget ($b = 1.0$) portfolio results on P0 F1 ($q^* = 10\%$), retained as an operational illustration rather than a validated method comparison. Paired differences are versus the better single-source baseline and are computed as material-level worst-source recall differences using a full-procedure grouped bootstrap: each replicate resamples reduced-formula groups with replacement, assigns distinct identities to duplicated occurrences, re-runs the portfolio strategy and both single-source baselines, and recomputes the improvement (see Note 1).

Strategy	Recall	NDCG	Coverage	ΔRecall	95% CI
JARVIS only	0.140	0.646	0.099	0.000	[0.000, 0.000]
MP only	0.140	0.673	0.099	0.000	[0.000, 0.000]
Maximin	0.386	0.909	0.099	0.246	[0.107, 0.286]
Balanced union	0.526	0.902	0.099	0.386	[0.273, 0.441]
Minimax oracle	0.474	0.915	0.099	0.333	[0.250, 0.414]

Table S4: Portfolio worst-source recall on P0 F1 across budget factors $b = 1.0, 1.5, 2.0$.

Strategy	$b = 1.0$	$b = 1.5$	$b = 2.0$
JARVIS only	0.140	0.281	0.333
MP only	0.140	0.228	0.316
Average percentile	0.456	0.526	0.579
Maximin	0.386	0.456	0.491
Balanced union	0.526	0.789	1.000

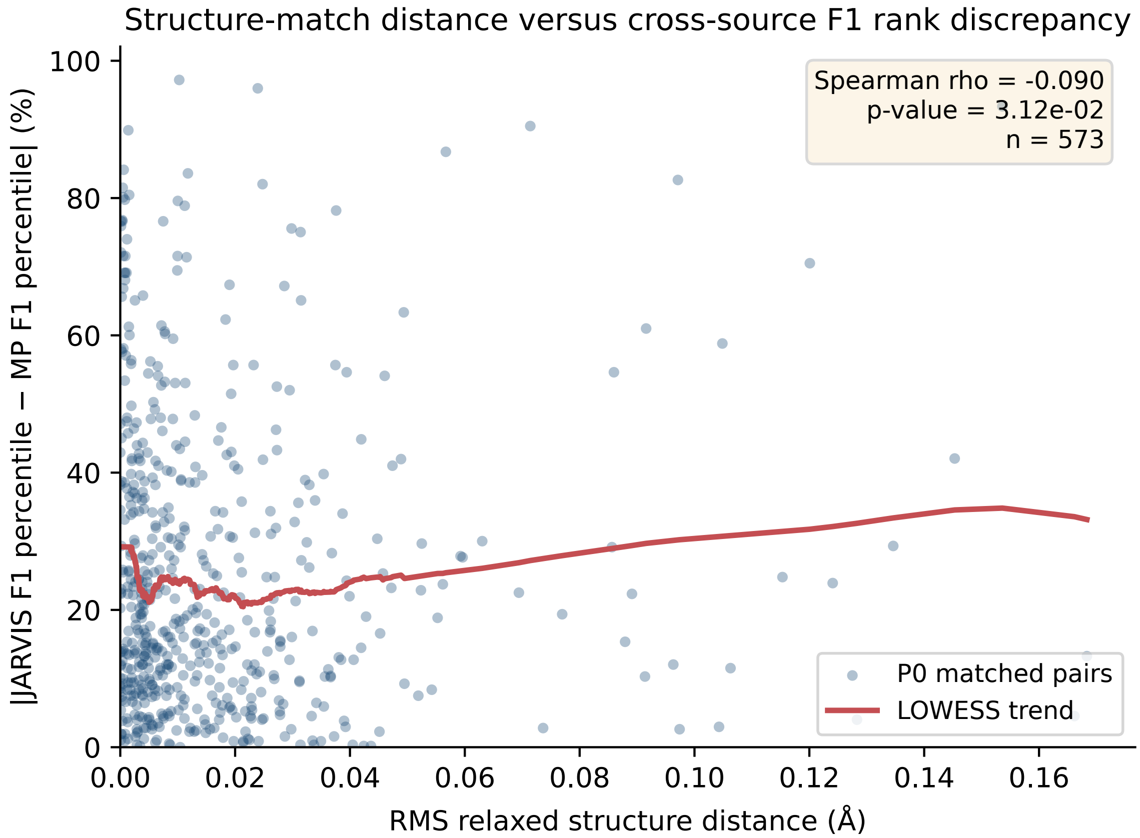

Figure S5: Relaxed-structure RMS distance versus absolute F1 percentile gap for the 573 P0 matched pairs. The association is statistically detectable at this panel size but negligible in magnitude ($\rho = -0.09$, $p = 0.0312$); this is a sensitivity diagnostic rather than an exclusion of threshold or matching-error effects. The LOWESS curve is illustrative and is not a fitted model.

Table S5: Baseline ranking diagnostics on P0 F1 (global Kendall $\tau = 0.245$, global Spearman ρ = 0.350). Columns are, from left to right: screened quantile q, corresponding top-k set size, plain Jaccard, chance-adjusted Jaccard $\widetilde{J}_q$, top-weighted Kendall τ on the union of the two top-k sets, top-weighted Spearman ρ on the same union, and the partial nAUCC of the band containing q (elite 1–10, intermediate 10–20, broad 20–50). The depth-agnostic $\mathrm{RBO}_{0.95}$ is shown once in Figure S4, rather than repeated in every row.

q (%)	k	Plain J	$\widetilde{J}_q$	τ_q	ρ_q	Band nAUCC
1	5	0.000	-0.005	-0.289	-0.588	0.003
5	28	0.018	-0.007	-0.532	-0.760	0.003
10	57	0.075	0.024	-0.386	-0.613	0.003
20	114	0.200	0.100	-0.331	-0.489	0.076
50	286	0.452	0.178	-0.080	-0.112	0.145

Table S6: Frozen structure-matching audit summary. P0 is the full relaxed-structure matched universe; P2 is the tighter subset retained by the RMS and lattice thresholds recorded in "panel_counts.json".

Panel	Count	Expected	RMS mean	RMS med.	RMS max	Max mean	Max med.	Max max	Lat. mean	Lat. med.	Lat. max
P0	573	573	0.017	0.009	0.168	0.025	0.013	0.298	0.066	0.004	1.897
P2	207	207	0.003	0.003	0.008	0.004	0.004	0.019	0.001	0.001	0.004

Table S7a: Overall screening-resolution results from the unified reduced-formula cluster bootstrap. Each of 2,000 replicates resamples reduced-formula groups, recomputes the bootstrap universe size and exact hypergeometric null, and yields a simultaneous studentized confidence band for the full curve. The nAUCC intervals are percentile intervals over the same replicates. The final column is the persistent practical-onset threshold $q_{0.05}^{\text{persist}}$, requiring five consecutive lower confidence bounds above 0.05.

Panel	Metric	N	Groups	τ [95% CI]	nAUCC [95% CI]	$q_{0.05}^{\text{persist}}$
P0	F1 Frobenius	573	496	0.245 [0.169, 0.318]	0.105 [0.056, 0.153]	–
P0	F3 Longitudinal	573	496	0.269 [0.195, 0.337]	0.110 [0.060, 0.157]	39%
P0	F4 Kelvin	573	496	0.255 [0.179, 0.329]	0.106 [0.059, 0.154]	42%
P2	F1 Frobenius	207	187	0.192 [0.058, 0.317]	0.060 [-0.028, 0.153]	–
P2	F3 Longitudinal	207	187	0.215 [0.083, 0.340]	0.043 [-0.032, 0.132]	–
P2	F4 Kelvin	207	187	0.205 [0.073, 0.331]	0.046 [-0.032, 0.140]	–

Table S7b: Banded partial nAUCC estimates for the elite (1–10%), intermediate (10–20%) and broad (20–50%) screening bands. Entries are estimates followed by percentile-bootstrap 95% intervals from the same reduced-formula cluster-bootstrap replicates; simultaneous coverage applies to the confidence band for the full screening-resolution curve.

Panel	Metric	Elite [95% CI]	Intermediate [95% CI]	Broad [95% CI]
P0	F1 Frobenius	0.003 [-0.023, 0.044]	0.076 [0.013, 0.141]	0.145 [0.087, 0.204]
P0	F3 Longitudinal	-0.000 [-0.023, 0.028]	0.061 [0.003, 0.127]	0.160 [0.096, 0.218]
P0	F4 Kelvin	-0.004 [-0.023, 0.038]	0.069 [0.010, 0.137]	0.151 [0.091, 0.209]
P2	F1 Frobenius	-0.023 [-0.030, 0.045]	0.030 [-0.048, 0.137]	0.095 [-0.032, 0.215]
P2	F3 Longitudinal	-0.021 [-0.031, 0.007]	-0.037 [-0.077, 0.053]	0.089 [-0.026, 0.211]
P2	F4 Kelvin	-0.021 [-0.031, 0.016]	-0.024 [-0.074, 0.093]	0.089 [-0.029, 0.210]

Table S8: Largest-RMS-distance trimming sensitivity on P0 F1. The primary matched panel is retained at 0%; the other rows remove the largest RMS distances and recompute the point-estimate resolution summaries.

Trim fraction	Removed count	Retained N	nAUCC	Elite partial nAUCC	Top-10 overlap
0%	0	573	0.105	0.003	8/57
1%	5	568	0.105	0.003	8/56
5%	28	545	0.104	0.005	8/54
10%	57	516	0.104	0.007	9/51
20%	114	459	0.094	0.001	8/45

Table S9: Tie-induced sensitivity of P0 control rankings. Exact ties were independently shuffled 1,000 times; ranges are tie-breaking sensitivity ranges, not sampling confidence intervals.

Attribute	Tied rows (J/M)	nAUCC range	Elite partial range	Top-10% shared-count range (out of 57)
Band gap	72/23	[0.897, 0.898]	[0.815, 0.815]	[51, 51]
Energy above hull	365/245	[0.532, 0.533]	[0.601, 0.601]	[43, 43]
Dielectric trace	4/17	[0.286, 0.286]	[0.197, 0.197]	[21, 21]